

TP MEGHANA SAI

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Professional Summary

AI and Data Science undergraduate with hands-on experience in machine learning and deep learning using Python. Worked on healthcare and medical datasets, handling data preprocessing, model training, and evaluation. Built and deployed ML models using TensorFlow and Scikit-learn for real-world applications. Strong interest in applying AI solutions to practical and data-driven problems. Comfortable working with structured data, model optimization, and basic deployment workflows.

Education

R.M.K College of Engineering and Technology

2022-Present

B.Tech in Artificial Intelligence and Data Science

CGPA : 8.04

Viswasai Junior College

2021-2022

12th - MPC

Percentage : 90.94

Akshara Vidyalaya

2020

10th - CBSE

Percentage : 79.2

Projects

Member Risk Stratification and Care Management

- Utilized Random Forest Regressor to forecast patient risk for the next 30, 60, and 90 days, assigning risk labels ranging from Very High Risk to Very Low Risk, achieving 93% accuracy.
- Developed a Flask backend for real-time predictions with seamless frontend integration and scalable healthcare risk assessment.

Handwritten Character Recognition using CNN on EMNIST Dataset

- Developed a CNN using TensorFlow to classify 62 EMNIST handwritten characters and digits, achieving over 98% test accuracy.
- Applied normalization, rotation correction, and optimized training with one-hot encoding and TensorFlow's tf.data pipeline.

Sepsis Prediction from Patient Health Data

- Built a Random Forest model using Python, Pandas, and Scikit-learn to predict sepsis onset from patient data, applying feature scaling and class-weight balancing to handle class imbalance.
- Achieved reliable diagnostic performance and deployed the model with joblib for real-time integration in healthcare systems.

Automatic Saline Level Monitoring System (IoT Based)

- Designed an IoT-based Automatic Saline Monitoring System using ESP8266 and load cell to track real-time IV fluid levels with remote alerts via Wi-Fi.
- Implemented web-based monitoring with Arduino IDE and Embedded C, enhancing patient safety and reducing manual intervention.

Technical Skills

- Programming Language:** Python
- Database and Cloud:** SQL, AWS
- Area of Interest:** Gen AI, Machine Learning, Deep Learning, Data Analytics & Visualization
- Data Analytics & Visualization:** Power BI, Excel
- Tools & Frameworks:** TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Certifications

- Oracle AI Vector Search Certified Professional**, Oracle(2024).
- Machine Learning A-Z™ Certification**, Udemy(August 2025).
- The Complete Python Bootcamp From Zero to Hero in Python**, Udemy(August 2025).
- Design Thinking(Elite silver)**, NPTEL(2024).

Languages

- English** (Fluent)
- Telugu** (Fluent)
- Hindi** (Fluent)
- Tamil** (Conversational)